

Analysis of warpage of a flip-chip BGA package under thermal loading: Finite element modelling and experimental validation

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ABSTRACT

The objective of this study is to investigate warpage of a Flip-Chip Ball Grid Array (FCBGA) package experimentally and numerically. Projection Moiré technique was used to measure full-field warpage during heating experiment from room temperature up to 250 °C. As temperature changes within the FCBGA package, different thermal strains are generated in the package resulting from the mismatch in the Coefficient of Thermal Expansion (CTE) between the constituent material layers. A finite element model and analytical equations based on the classical bending theory were used to analyze the thermally induced deformations of package under test. The results show that a maximum warpage value of 67 µm with a concave or smiling shape occurred at the neighborhood of 70 °C during heating and 87 µm warpage with a convex crying shape at 250 °C by assuming a stress-free temperature at 175 °C. Warpage analysis showed that the studied FCBGA package exhibits a different curvature in the die region at the center of the package compared to the region outside the die. At temperatures higher than 175 °C, a concave or smiling warpage shape was shown in the die region while a convex or crying shape in the region outside the die. In contrast, at room temperature, the die area has a convex shape warpage, while outside this area presented a concave shape.

Furthermore, the good agreement between warpage of the studied package from finite element simulation and measurements was achieved thanks to the use of the measured material's thermo-mechanical properties of the substrate and Molding Compound (MC) rather than using datasheet data. In-plane CTE and Young's modulus of the substrate were measured using stereo-digital image correlation. Moreover, temperature dependent Young's modulus of the MC was measured by nanoindentation.

1. Introduction

Flip-chip interconnect technology has been widely used for packaging semiconductor devices owing to its small package size, high I/O density, as well as thermal and electronic performances [1,2]. However, warpage of Flip-Chip Ball Grid Array (FCBGA) packages is one of the critical challenges. The large mismatch in the Coefficient of Thermal Expansion (CTE) between the chip, the substrate and the Molding Compound (MC) is one of the root causes for reliability issues like large warpage, cracking or delamination in the Back End Of Line (BEOL) [3,4] and bump cracking during the FCBGA manufacturing, functioning, or reliability tests, as they are subjected to temperature changes or gradients of temperature. Among these challenges, this paper will focus on studying warpage of a FCBGA package. The measurement and prediction of warpage behavior in FCBGA packages is crucial to insure a good mounting to the printed circuit board.

Up to now, some research studies has already been conducted in the last several years to investigate warpage of integrated circuit packages. These studies consist of exploring various experimental and numerical techniques [1,5,6]. Warpage values from Finite Element (FE) simulations or analytical models of electronic packages were compared to experimental ones measured by using techniques like shadow Moiré and projection Moiré [7].

In order to accurately predict warpage of integrated circuit packages using FE simulations or analytical models, accurate measurements of temperature dependent thermo-mechanical properties of the materials used to make the FCBGA packages are required. Many studies highlighted the importance of material's characterization in predicting warpage [1,5,8,9]. Tsai et al. [1,5] used Dynamic Mechanical Analysis (DMA) and Thermal Mechanical Analysis (TMA) to measure the temperature dependent Young's modulus and CTE of the MC and substrate materials in their warpage study of molded FCBGA package [1] and bare

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die FCBGA [5]. Zhang et al. [8] combined DMA, curvature measurement of a bi-layer of MC-substrate materials and Stoney equation, in order to measure the temperature dependent CTE and Young's modulus of MC and substrate materials for their warpage study of a bi-layer beam sample. De Vreugd et al. [9] used DMA to characterize the viscoelastic behavior of the MC. The properties of the MC were used to predict the deformed shape of a bi-layer of MC and silicon at room temperature as well as for a Quad Flat No-leads (QFN) panel.

In the reported warpage studies of integrated circuit packages, warpage values from FE simulation or analytical models were compared to experimental ones. De Vreugd et al. [9] showed a good agreement between the FE and experiment when comparing the shape of a double layered beam MC-silicon at room temperature. Tsai et al. [1] in their study of warpage of bare die FCBGA packages also showed a good agreement between FE simulation and shadow moiré measurements at different temperatures. The deformed shape of the package from experiment and simulation showed also a quite good agreement, but the comparison was conducted only for two temperatures.

Up to now, none of the reported research have studied in detail the shape of a molded FCBGA packages at different temperatures and the difference in curvatures within the package. In the current study, in addition to the warpage study of a molded FCBGA package, extensive analysis will be conducted to thoroughly understand curvature and warpage changes at different temperatures and the link to the properties of the different materials present in the package. To that end, an analytical solution, presented in the following section, will be used.

In Section 3, the studied FCBGA package is presented. Section 4 is devoted to material's characterization. Two different techniques were used to measure the overall properties of the package substrate and the MC material as these two play an important role in the warpage of molded FCBGA packages. Nanoindentation [10,11] was used to measure the Young's modulus of the MC material at different temperatures. The advantage of this technique is that it does not need a standalone material to test. Furthermore, Stereo-Digital Image Correlation (SDIC) technique [12,13] was used to measure the overall in-plane CTE and Young's modulus of the studied FCBGA package substrate. The SDIC technique allows contactless and full-field displacement and strain measurements of materials. This technique is suitable for CTE measurement and Young's modulus measurements of thin film samples. One of the advantages of this technique is the ability to capture the in-plane anisotropy of the tested material with a single test compared to strain gauges that can measure only local and unidirectional strains.

Section 4 and 5 present the tool setup used to measure warpage of the studied FCBGA package and the used FE model for warpage predictions. The measured thermo-mechanical properties of the MC material and the substrate were used as input to FE simulations. Finally, Section 6 is devoted to the analysis of warpage and curvature changes at different temperatures in the die area as well as in the area outside the die.

2. Analytical solution for multi-layer structures

The analytical equations, presented in this section, are based on the classical bending theory [14], assuming the individual force and bending moment in each layer. The bending moment is related to the curvature of the layer, and all layers are assumed to have the same curvature. The solution is obtained by balancing the forces and moments in the system and satisfying the strain continuity condition at the interfaces between the different layers [15,16].

Let's consider a plate consisting of n bonded thin-film layers as depicted in Fig. 1. The plate is subjected to internal strains resulting from their deposition. The strains are assumed isotropic and uniform. Such strains generate biaxial stresses in each layer:

$$\sigma_i(z) = E'_i \varepsilon_i(z) \quad (1)$$

where $E'_i = E_i/(1 - \nu_i)$ is the biaxial modulus, E_i is the Young's modulus

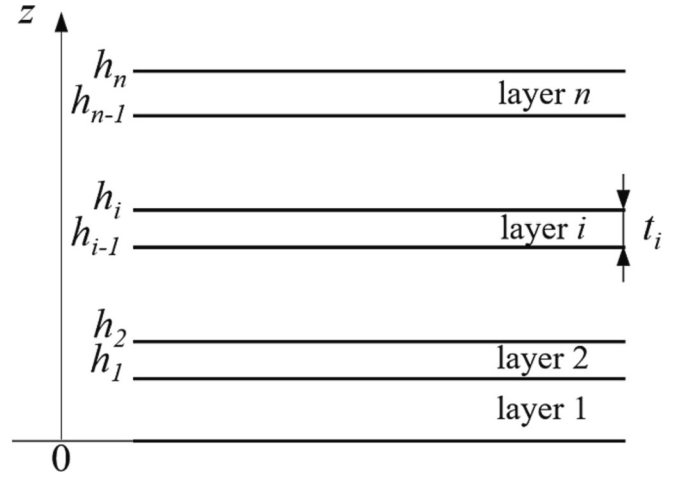


Fig. 1. Schematics showing a cross-section of a multilayer plate.

and ν_i is the Poisson's ratio of the layer i .

These stresses lead to biaxial forces and internal moments which cause the structure to bend in addition to planar contraction/expansion. The equilibrium condition can be written:

$$\sum_{i=1}^n \int_{h_{i-1}}^{h_i} \sigma_i(z) dz = 0 \quad (2)$$

$$\sum_{i=1}^n \int_{h_{i-1}}^{h_i} \sigma_i(z) z dz = 0 \quad (3)$$

where h_{i-1} and h_i are the z coordinates of the lower plane and upper plane of the layer i respectively.

Eq. (2) leads to the biaxial elastic strain ε_i^F :

$$\varepsilon_i^F = \frac{\sum_{i=1}^n E'_i t_i \varepsilon_i^0}{\sum_{i=1}^n E'_i t_i} - \varepsilon_i^0 = \bar{\varepsilon}_i^0 - \varepsilon_i^0 \quad (4)$$

where $\bar{\varepsilon}_i^0$ is the weighted average strain of the different layers.

The residual biaxial stress in each layer originates from the planar contribution resulting from the strain mismatch $\varepsilon_i^0 - \bar{\varepsilon}_i^0$ plus the bending contribution emanating from the curvature K . The bending contribution depends linearly on the coordinate z and vanishes in the neutral bending plane (h_n), can be written as:

$$\varepsilon_i^M = K(z - h_n) \quad (5)$$

Combining Eqs. (3) and (5), h_n can be written:

$$h_n = \frac{\sum_{i=1}^n E'_i t_i h_{mi}}{\sum_{i=1}^n E'_i t_i} \quad (6)$$

where h_{mi} is the mid-plane height:

$$h_{mi} = \frac{h_i + h_{i-1}}{2} = h_i - \frac{t_i}{2} \quad (7)$$

Combining eqs. (1) and (3-7), the curvature K can be formulated as:

$$K = \frac{\sum_{i=1}^n E'_i t_i [h_{mi} - h_n] (\varepsilon_i^0 - \bar{\varepsilon}_i^0)}{\sum_{i=1}^n E'_i t_i \left[h_i^2 + \frac{t_i^2}{3} - h_i t_i + h_n (h_n - 2h_{mi}) \right]} \quad (8)$$

3. Description of the studied FCBGA package

The studied package is FCBGA package with 0.5 mm pitch [17–19]. The test chip has 100 μm thickness and $10 \times 10 \text{ mm}^2$ in size. It contains area array of copper pillar at a pitch of 160 μm . The diameter and height of each pillar are 50 μm and 35 μm , respectively. On top of each copper pillar there is 15 μm thick solder cap. The package uses a 400 μm thick substrate with 6 copper layers. The test chip is attached to the substrate using a Thermo-Compression Bonding (TCB) of copper pillar flip-chip with a Non-Conductive Paste (NCP) as pre-applied underfill [20].

First, the NCP was dispensed onto the substrate prior to TCB process. Then, the Flip-Chip (FC) was placed onto the dispensed NCP and this latter was spread uniformly below the chip. The FC was compressed and solder cap of the copper pillar flattened to make area for solder wetting on each pillar. The chip was heated up and cooled down following a temperature profile, so the solder joints were formed on the bond pads and the NCP was cured on the bonding interface. After TCB, the FCBGA was over-molded and the MC material was cured at 175 $^{\circ}\text{C}$. Fig. 2.a and c depicts a schematic cross-section as well as a real cross-section of the studied package, showing the area delimited by red dashed lines. Fig. 2. b presents an image of the $19 \times 19 \text{ mm}^2$ molded package. Table 1 summarizes the dimensions of the package.

4. Material's characterization

4.1. Molding compound

The molding compound is a heterogeneous material, made of a mixture of different materials among them inclusions embedded in a polymer resin type material such as epoxy. Its thermo-mechanical properties depend on the processing conditions, geometry, confining conditions, as well as temperature. Young's modulus of the MC material is known to be temperature dependent. So, having this information is quite important for the reliability assessment of FCBGA packages.

Table 1

Geometrical properties of the FCBGA package.

Package size	$19 \times 19 \text{ mm}^2$
Chip size	$10 \times 10 \text{ mm}^2$
Substrate thickness	400 μm
MC material thickness	450 μm
chip thickness	100 μm
Pillars pitch	160 μm
Ball pitch	500 μm

Nanoindentation tests were performed to measure the temperature-dependent Young's modulus the MC material using a TI950 Triboindenter from Bruker® (see Fig. 3.a). This tool is equipped with a xSol-600 heating stage (cf. Fig. 3.b), which is constructed with low thermal expansion and thermally insulating materials to achieve high thermal stability. It includes dual heaters controlled by independent Proportional Integral Derivative (PID) controller to heat the top and bottom of the sample. Its design allows a good temperature control and uniformity.

A standard Berkovich diamond indenter probe was used to perform quasi-static indentation experiments. As the indenter tip is pressed into the sample, both the depth of penetration h and the applied load P are monitored. From the experimentally obtained load versus depth curve, the Young's modulus E is calculated using the Oliver and Pharr method [10,11], in which the measured contact stiffness S is given by:

$$S = \beta \frac{2}{\sqrt{\pi}} E_r \sqrt{A} \quad (9)$$

where A is the projected contact area of the indenter with the sample surface, β is a dimensionless parameter close to unity that accounts for deviations in stiffness caused by the lack of axial symmetry for pyramidal indenters, and E_r is the effective Young's modulus, defined by

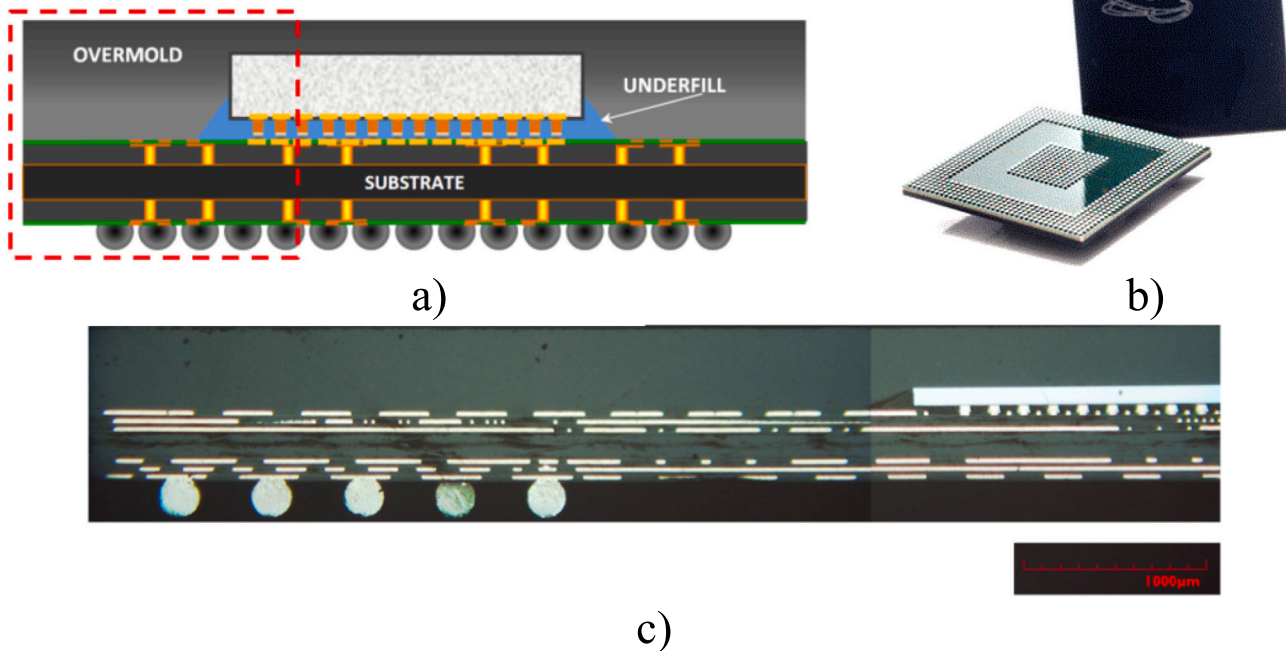


Fig. 2. The studied FCBGA package: a) schematic cross section of the package, b) top and bottom views of the package, c) real cross section of the package showing the area delimited with red dashed line in a. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

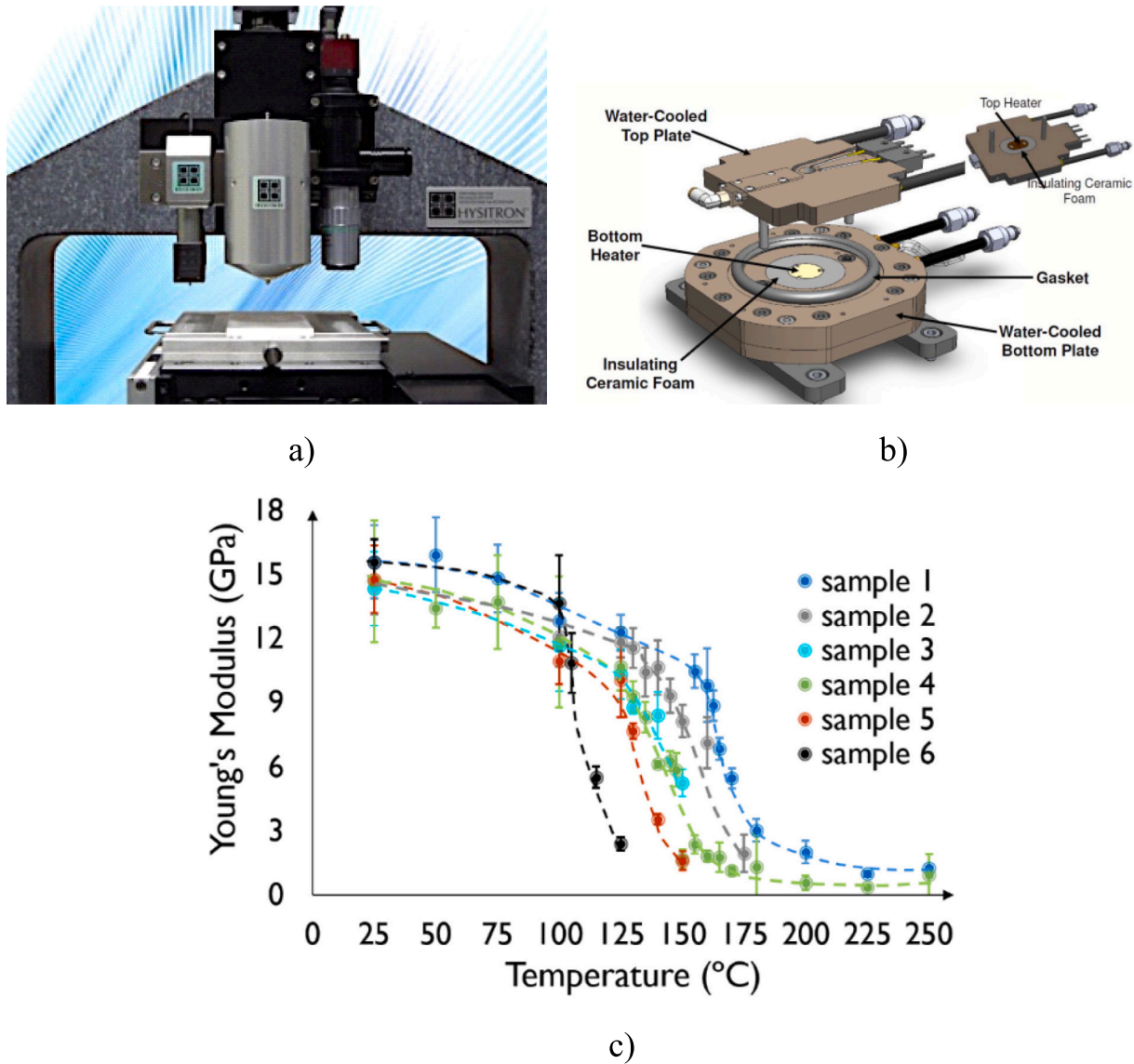


Fig. 3. Nanoindentation test set-up and results: a) TI950 Triboindenter tool from Bruker®, b) xSOL-600 heating stage, d) measured Young's modulus for six tested samples at different temperatures ranging from room temperature up to 250 °C.

$$E_r = \left[\frac{1 - \nu^2}{E} + \frac{1 - \nu_i^2}{E_i} \right]^{-1} \quad (10)$$

The effective Young's modulus takes into account the elastic displacement occurring in both the sample, with Young's modulus E and Poisson's ratio ν , and the indenter probe, with elastic constants, $E_i = 1140$ GPa and $\nu_i = 0.07$.

The measurements were performed at different temperatures ranging from room temperature up to 250 °C. Since the MC material is a mixture of different type of inhomogeneous materials, for each measurement, at least, five tests were carried-out at different locations and the average value of the measured Young's modulus was considered. Fig. 3.d presents the curves Young's modulus results vs. temperature for six tested samples. As expected, the effective Young's modulus decreases as the temperature increases. A large reduction of the Young's modulus about 7.5GPa is observed in the range of temperature $T_g \pm 12$ °C (glass transition region). Then, it becomes almost constant again in the rubbery plateau. In addition, the error bar indicates the standard deviation. The standard deviation is 12 to 20 times higher at room temperature compared to its value at 250 °C (1.2–2.1 GPa at 25 °C and around 0.1 GPa at 250 °C).

From Fig. 3.d, a large variability from sample to sample is observed, probably due to different processing conditions. A large variability in the glass transition temperature T_g value is observed as well (115 °C–170 °C). The Young's modulus of the MC material under test varies from 14 to 15.6 GPa at room temperature and from 930 MPa to 1.24 GPa at 250 °C.

4.2. Substrate

The studied package substrate is 0.4 mm thick and has a symmetric stack. It is made of 6 electrically conductive copper layers and insulating core and prepreg layers in between as shown in cross-section in Fig. 4. The central layer is made of the core material and consists of a composite material with glass fibers and epoxy resin. The prepreg layers are similar to the core material but thinner. A solder mask layer is applied on top and bottom of the substrate to protect the copper traces from oxidation. Since these materials have different thermo-mechanical properties (cf. Fig. 4) and copper lines and vias are present with different densities among the different layers, the modelling of such a structure is complex and time consuming. Thus, overall properties of the substrate will be measured and used as input for FE simulations.

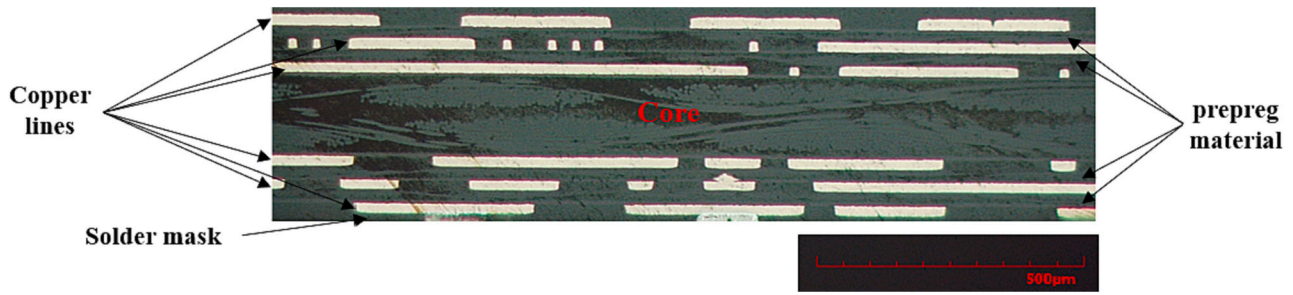


Fig. 4. Cross-section of the studied substrate with 6 copper layers.

In this part, the CTE and Young's modulus of the used substrate was measured using SDIC technique [12,13]. The samples have copper pads pattern on their surface that can be used in the SDIC analysis. The out-of-plane deformation of the substrate is neglected as it has a symmetric stack layer in the thickness direction.

4.2.1. CTE measurements

In one of our previous papers [21], we used two-dimensional Digital Image Correlation (2D-DIC) to measure the CTE of pure copper as a reference to check the feasibility and effectiveness of the methodology. The measured CTE was in good agreement with the values from the literature. In the current study, SDIC was used instead of standard 2-D DIC. For this latter technique, the specimen must be positioned parallel to the camera and must undergo a planar deformation without any out-of-plane displacement, which is difficult to guarantee. The same

methodology as in [21] was followed to measure the in-plane CTE of our package substrate.

The used experimental setup for the in-plane CTE measurements is displayed on Fig. 5.a. It consists of a SDIC system from Correlated solutions and a heating stage used to heat the sample under analysis. The SDIC system has two synchronized monochromatic cameras with 5-megapixel resolution and equipped with two 135 mm Rodagon® lenses. Two white light sources are used to illuminate the Region of Interest (ROI). The samples were covered by a black paint pattern speckle on top of a thin white spray.

The SDIC system was calibrated using a calibration plate (cf. Fig. 5. b). Images of the calibration plate were acquired at different positions. The plate needs to be translated, tilted and rotated. The calibration plate pairs of images were used to determine the calibration parameters for each camera as well as the 3-D relative position/orientation of the two

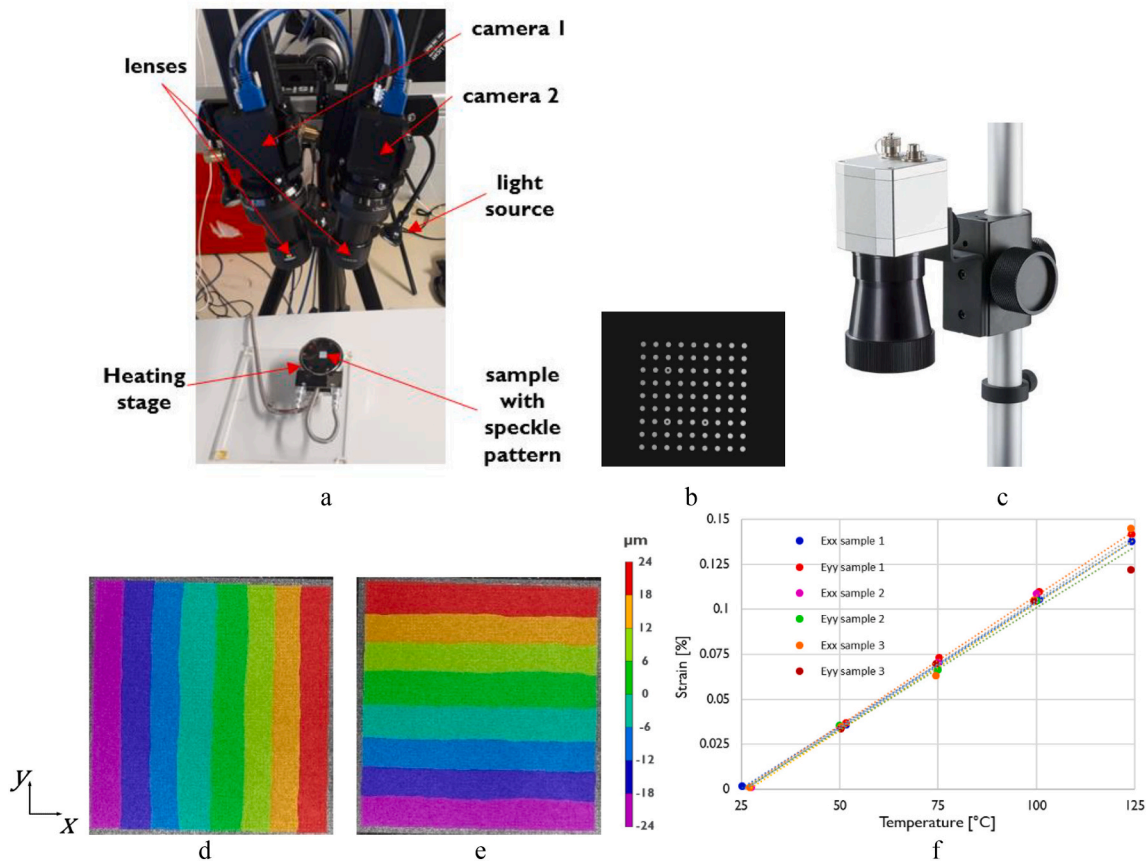


Fig. 5. CTE measurement of the substrate: a) SDIC experimental system set-up from correlated solutions® equipped with a heating stage, b) calibration target, c) Optiris® PI450 Infra-Red camera for temperature measurements. d and e show the measured in-plane displacement fields from SDIC analysis of the studied substrate expansion between room temperature and 125 °C in the in-plane directions x and y respectively. f) average thermal expansion strains ϵ_{xx} and ϵ_{yy} versus temperature for three substrate samples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

cameras. Lens distortions are also corrected during calibration for a better measurement accuracy. After determining the calibration parameters, the 3-D specimen shape can be reconstructed using stereo-triangulation.

Pairs of images of the ROI were captured at different temperatures going from room temperature to 125 °C. For each temperature step, a stabilization time was necessary to achieve the thermal equilibrium. The temperature was measured using an Optris® PI450 Infra-Red camera with a spectral range of 7.5–13 µm and 382 × 288 pixel resolution, equipped with a 10° × 8°, $f = 44$ mm lens (Fig. 5.c). Mean temperatures over the ROI was considered for each temperature step. 35 pairs of images were recorded at each stabilized temperature. The pairs of images acquired before and after deformation were analyzed, using the VIC3D® software [13], to calculate the in-plane displacements and strains induced by temperature change. Each sequence of 35 images was then processed numerically to retrieve the average displacement fields in the time domain corresponding for each temperature.

Fig. 5.d-e illustrates the uniform thermal expansion of the substrate sample, when temperature increases from room temperature up to 125 °C, as the in-plane displacements fields in x and y directions are shown to be linear depending on the coordinates x and y . The average in-plane strains from the region of interest are then extracted for the calculation of the coefficient of thermal expansion (CTE) α given by the equation:

$$\alpha = \frac{\varepsilon}{\Delta T} \quad (11)$$

where: ε is the average in-plane strain due to the temperature increase ΔT .

The average strains ε_{xx} and ε_{yy} as a function of temperature for the three measured samples is presented in Fig. 5.f. It can be seen from this figure that the thermal expansion increases approximately linearly with the increase of temperature. The CTE of the measured samples was determined by fitting the data using straight lines. The measured CTE data are presented in Table 2. It is observed that the CTE values in the x direction and y direction are quite similar. The average measured CTE value will be considered for the finite element simulation. The measured value is a bit lower than the CTE of pure copper. The copper plays an important role in the thermal expansion as it is more rigid than the core and the prepreg materials. The influence of the solder mask can be neglected as it is too thin compared to the other constitutive materials. The copper density plays an important role in the optimization of substrate CTE.

4.2.2. Young's modulus measurements

The experimental tool setup used to measure Young's modulus is shown in Fig. 6.a. This setup is similar to the one used for CTE measurements in which the heating stage is replaced by a Deben® tensile test apparatus used to apply tensile loads to the substrate stripe samples. A particular attention is paid to obtaining the best possible alignment between the sample axis and the tensile test machine loading axis so as to minimize the parasitic bending cause by the misalignment. Cyclic loading/unloading tests were performed at room temperature using a crosshead velocity of 0.05 mm/min. The elastic modulus was calculated from the linear portion of the stress-strain curve during unloading, where it is expected that the specimen undergo only elastic deformation. Load-displacement from the tensile test apparatus data were streamed to a computer (see Fig. 6.b). The full-field in-plane displacements and

strains were calculated using SDIC by analyzing the recorded pairs of images of a rectangular region centered on the upper face of the tested sample and recorded at different loading stages. Fig. 6.c-d shows linear displacements fields in longitudinal and transverse directions respectively, resulting from force unloading between the two stages: "image 1" and "image 2" (see Fig. 6.b). Consequently, the strain is quite uniform in the area of interest. The average longitudinal and transverse strains from the region of interest are extracted for Young's modulus E and Poisson's ratio ν calculation using the equations:

$$E = \frac{\Delta F/A}{\varepsilon_{xx}} \quad (12)$$

$$\nu = \frac{\varepsilon_{yy}}{\varepsilon_{xx}} \quad (13)$$

where: ΔF is the change in measured load from the tensile testing machine, A is the initial cross-sectional area of the sample and ε_{xx} is the average longitudinal strain and ε_{yy} is the average transverse strain between two loading stages calculated using SDIC. The obtained mechanical properties for 3 samples are summarized in Table 3. For the finite element simulation, the average measured properties will be considered.

5. Warpage measurement

A Topography and Deformation Measurement (TDM) tool from Insidix® [22], displayed in Fig. 7.a, is used to measure the out-of-plane deformation of FCBGA packages under investigation in the temperature range going from room temperature up to 250 °C. The TDM uses the projection Moiré technique, which is a non-contact and full field method. A video projector illuminates the sample surface with light stripe patterns consisting of equidistant parallel lines. The resulting image is recorded by a Charge-Coupled Device (CCD) camera. On perfectly flat and horizontal surfaces, the light pattern is a set of equidistant, parallel stripes. Any deviation of the sample surface from flatness will lead to a deformation of the stripe patterns. The amount of deformation of the stripe patterns is related to the deformation or change in height of the surface under investigation.

In addition, TDM is equipped with infra-red heating lamps and thermocouples that can be attached to the FCBGA surface used to control the temperature. This allows topography measurements at different temperatures. For each measurement, the temperature is kept constant. Fig. 7.b shows, as example, the measured MC surface topography of one FCBGA package at room temperature.

6. FE modelling

The FCBGA package is considered stress free at 175 °C. The finite element model was built using the software MSC Marc® [23], considering the 1/8 symmetry of the FCBGA package (cf. Fig. 8). Details of the copper pillars, the underfill, the BEOL and the substrate were omitted in this model since we are interested in the global deformation of the FCBGA package. Therefore, equivalent orthotropic thermo-mechanical properties were calculated for copper pillars and underfill and the BEOL using the concept of Representative Volume Element (RVE). For the substrate, the measured overall CTE and Young's modulus were used as input to the FE simulation. Material's properties used to package the FCBGA are listed in Table 4. In this model, all the materials were assumed linear elastic. It is important to note that temperature dependent properties were considered for the underfill and the MC material taking into account the glass transition temperature (T_g). The BEOL equivalent properties calculated by Gonzalez et al. [18] are summarized in Table 5, however, the obtained equivalent properties for copper pillars and underfill are listed in Table 6.

Table 2
Measured CTE values of the studied substrate.

	CTE _x [ppm/°C]	CTE _y [ppm/°C]
Sample 1	14.44 ± 0.80	13.97 ± 0.59
Sample 2	14.19 ± 0.96	14.11 ± 1.09
Sample 3	14.35 ± 1.03	14.08 ± 1.20

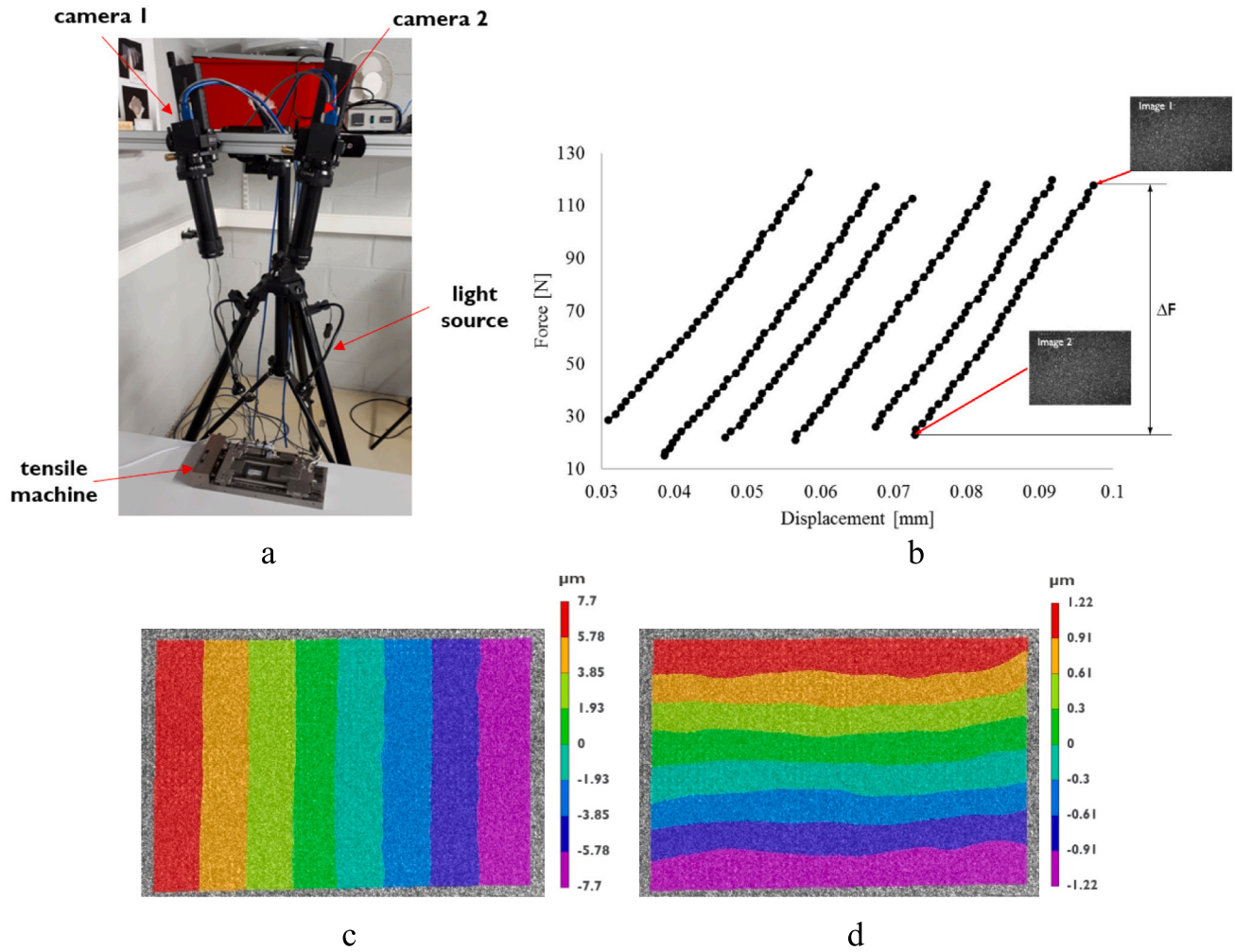


Fig. 6. Young's modulus measurement of the substrate: a) SDIC experimental system set-up from correlated solutions® equipped with a Deben® tensile testing machine, b) Force-displacement curve from a tensile test (6 loadings-unloadings). c and d show the measured in-plane displacement fields from SDIC analysis of the studied substrate during unloading in longitudinal and transverse directions respectively.

Table 3

Measured Young's modulus values of the studied substrate.

	E [GPa]	ν
Sample 1	31.36 ± 0.35	0.238 ± 0.019
Sample 2	31.97 ± 0.47	0.246 ± 0.012
Sample 3	31.17 ± 0.82	0.238 ± 0.015
Average	31.80	0.240

7. Results and discussion

The FCBGA package can be divided in two regions with different cross-sections: region 1 is the central portion of the FCBGA package within the die perimeter. It contains the substrate, pillars and underfill, die and MC layers (cf. Fig. 8.c). Region 2 is the package portion outside the die area when looking from top as depicted in Fig. 8.d. It includes the substrate and MC layers as shown in Fig. 8.c. It is expected that the two regions will exhibit different curvatures. In fact, when cooling or heating the package, the different layers will contract or expand with different thermal strains due to their different CTE. Thereby, the mismatch in thermal strains between these layers results in an in-plane tension/compression plus a bending since the different layers adhere to each other.

Profiles are extracted along the two diagonals for each temperature (cf. Fig. 9.a). The figure indicates the convention for sign of warpage and

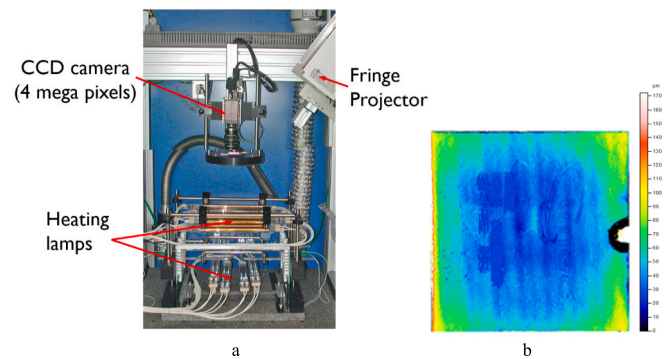


Fig. 7. a) Overview of the topography measurement TDM Insidix® setup. b) Topography measurement result of MC material surface of one FCBGA package under study at room temperature.

curvature: a positive value is given to a concave or a crying shape warpage and a negative value for a convex or smiling shape warpage. From Fig. 9.a, at temperatures above 175 °C, a concave shape in region 1 at the center of the FCBGA package is observed and a convex shape in region 2. In contrast, at 25 °C, region 1 has a convex shape warpage while region 2 has a concave shape.

The difference between maximum and minimum amplitude of the

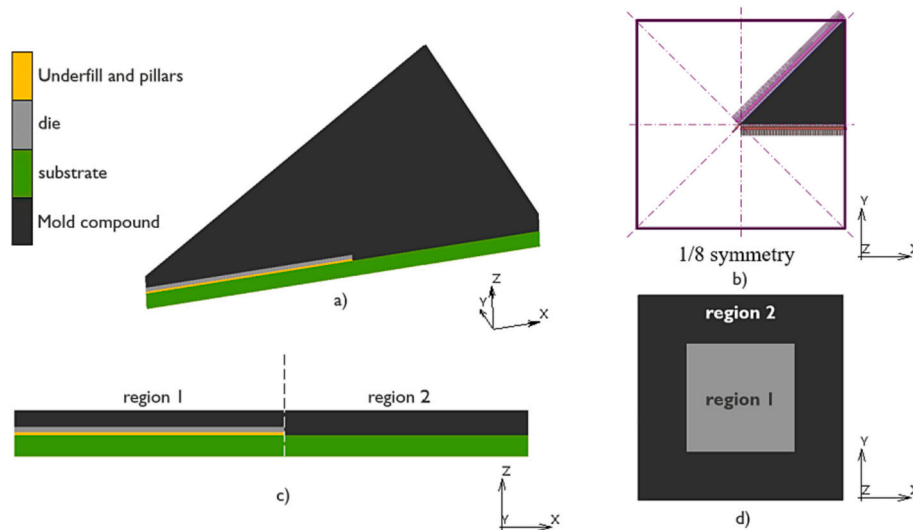


Fig. 8. Three-dimensional Finite Element Model of the studied FCBGA package considering 1/8 symmetry as shown in a). b) shows the eight symmetry boundary conditions. c) presents a cross sectional view of FCBGA package divided in two regions region1 and region2 for deformation analysis. d) is top view showing region1 delimited with the die size and region 2 outside the die area (MC on top of the die was removed).

Table 4
Thermo-mechanical properties used in the FE simulation.

	Tg [°C]	CTE [ppm/°C]		Young's Modulus [GPa]		Poisson's ratio []
		(<Tg)	(>Tg)	(<Tg)	(>Tg)	
copper		17		117		0.34
underfill	100	26	90	9	0.08	0.3
Equivalent substrate	–	14.2		31.8		0.2
Si	–	2.3		169		0.26
MC material	130	14	70	Fig. 17		0.3

two diagonal profiles is considered as the package warpage. In this way, warpage values are obtained at different temperatures. The experimentally obtained warpage values of 6 FCBGA packages, during heating from 25 °C up to 250 °C, are compared to the calibrated finite element model results by fine tuning the MC material Young's modulus (cf. Fig. 9.b). A good agreement is achieved between FE and experiment. Moreover, the obtained MC material Young's modulus is consistent with the Young's modulus measurements (cf. Fig. 9.c). From this figure, a large variability in warpage measurements from package to package is observed (ranging from 17.5 to 28 μm), which can be linked to the variability in the Young's modulus measurements of the MC material observed in Fig. 9.c.

From Fig. 9.b, FCBGA package warpage is around -50 μm (smiling

shape warpage), it increases to reach a warpage value of -67 μm after heating up to around 70 °C. At temperatures between 70 °C and 115 °C, warpage is almost constant. Upon further heating, warpage value decreases and reach zero at around 175 °C, which was considered the stress-free temperature. For temperatures higher than 175 °C, the package becomes convex (smiling shape warpage). Its warpage value continues increasing with temperature increase to reach a value around 87 μm at 250 °C.

The analytical equations presented in Section 2 were used to analyze the deformation in region 1 and 2 for the sake of understanding its root cause. The analysis of the deformation in region 2 outside the die, is more straightforward, since only MC and substrate layers are present in the stack. According to Fig. 10.b, when the temperature increases from stress-free temperature (175 °C) to 250 °C, MC material expands more than 4 times than the average thermal strain of the stack MC-substrate, while the substrate expands 1.14 times less. Hence, MC is in compression while the substrate is in tension, and since the MC mid-plane is on top of the neutral bending plane (cf. Fig. 10.h), the moment generated by the two layers are in the same direction inducing a concave shape warpage at temperature between 175 °C and 250 °C. By contrast, the deformation has opposite sign in the temperature range between 25 °C and 120 °C inducing a convex warpage shape.

As shown in Fig. 10.g, region 2 exhibits a low curvature (1.7 mm^{-1}) between 175 °C and 250 °C because of the low flexural rigidity of the MC on one hand ($\approx 6 \text{ N.mm}$) compared to the substrate's rigidity which is around 176 N.mm . On the other hand, the substrate has a low bending

Table 5
Equivalent orthotropic thermo-mechanical properties obtained for the equivalent BEOL [18].

Young's modulus [GPa]			Shear modulus [GPa]			Poisson's ratio []			CTE [ppm/°C]		
E ₁₁	E ₂₂	E ₃₃	G ₁₂	G ₂₃	G ₃₁	ν_{12}	ν_{23}	ν_{31}	α_{11}	α_{22}	α_{33}
99.87	101.63	46.41	38.13	32.51	10.92	0.224	0.185	0.086	4.53	4.73	15.84

Table 6
Equivalent orthotropic thermo-mechanical properties obtained for the equivalent underfill and copper pillars.

	Young's modulus [GPa]			Shear modulus [GPa]			Poisson's ratio []			CTE [ppm/°C]		
	E ₁₁	E ₂₂	E ₃₃	G ₁₂	G ₂₃	G ₃₁	ν_{12}	ν_{23}	ν_{31}	α_{11}	α_{22}	α_{33}
(<Tg)	10.87	10.87	11.60	3.26	3.14	4.63	0.30	0.28	0.32	25.9	25.9	21.2
(>Tg)	0.11	0.11	0.29	0.03	0.03	0.12	0.30	0.02	0.32	99.8	99.8	21.3

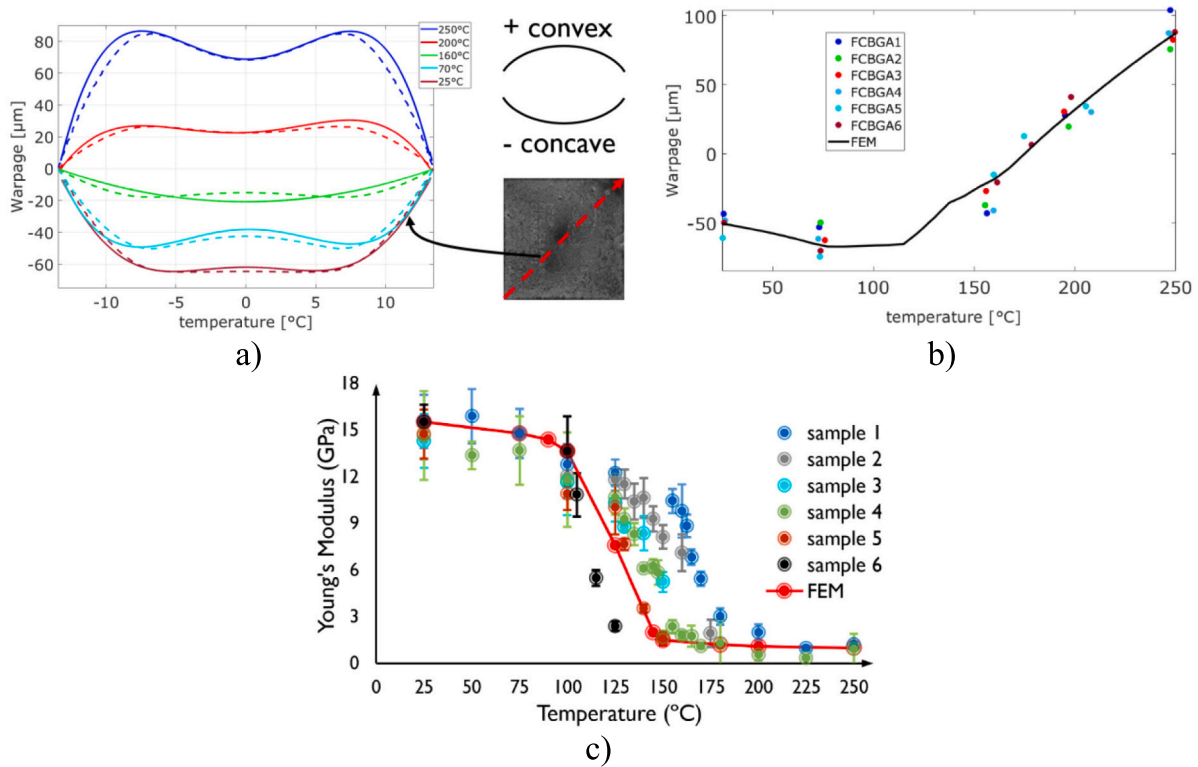


Fig. 9. Correlation between measurement and Finite Element Model (FEM) results: a) diagonal profiles of MC material surface of one studied FCBGA package at different temperatures: measured in solid lines, FEM results in dashed lines. b) Measured warpage for 3 FCBGA packages vs. calibrated FEM model results at different temperatures. c) calibrated Young's modulus vs. temperature of the MC material compared to the measured values by nano-indentation.

contribution since the distance between its mid-plane to the neutral bending plane is in the range of 14–18 μm (cf. Fig. 10.h). However, when the temperature goes below 175 °C, MC layer gets more rigid while cooling down through glass transition temperature from the rubbery state towards the glassy state (Young's modulus increases by 12 times from 175 °C to 25 °C). At the same time the neutral bending plane moves up towards the MC layer making the substrate's contribution to bending higher.

When cooling down from 90 °C to 25 °C, the difference between the thermal strains induced by the MC and the average axial strain of the whole stack substrate-MC decreases (cf. Fig. 10.d). This is due to the larger increase of the average thermal strain compared to the thermal strains induced by MC shrinkage as shown in Fig. 10.b. This results in a curvature reduction in region 2 in that range of temperatures (cf. Fig. 10.g).

In region 1, in the center of the package, the cross-section contains the substrate, the pillars and underfill, the BEOL, the die and the MC layers. As, the BEOL is only 3 μm thick, its flexural rigidity is about 2×10^{-4} N.mm and hence its contribution to the package bending is negligible. In order to further simplify the problem, the pillars and underfill contribution to the package bending could also be neglected as their mid-plane is close to the neutral bending plane (see Fig. 10.e).

The MC and the substrate layers deform more than the average axial deformation of the whole stack substrate-pillars-die-MC; however, the die deforms less (cf. Fig. 10.c). The die and MC mid-planes are above the neutral bending plane (cf. Fig. 10.h). As a consequence, both the substrate and die layers contributions to bending of the FCBGA package in region 1 is in the same direction, while the MC layer acts in the opposite direction (cf. Fig. 10.e). The sum of bending contributions from the die and substrate are larger than MC contribution. Thus, the sign of the package curvature is given by curvature of the die and substrate. At temperatures higher than 175 °C, the substrate and die contributions to the bending of the package are larger than those of MC which makes the

center of the package concave. When cooling down from 175 °C to 138 °C, region 1 deforms in the opposite direction and hence the center of the package is convex. When cooling down from 138 °C and 110 °C, the flexural rigidity of the MC decreases by 4 times, due to the glass temperature transition. Hence, MC induces a larger increase in bending contribution compared to the substrate's and die's contributions. Therefore, the curvature of the whole stack reduces in region 1. As the temperature continues to decrease from 110 °C to 25 °C, the increase in flexural rigidity of the MC is lower and the neutral bending plane continues to move up increasing the bending contribution of the substrate. The bending contribution of the substrate is larger inducing an increase in the curvature towards a more convex shape in the center of the package.

8. Concluding remarks

The warpage of six FCBGA packages under thermal loading from room temperature to 250 °C was measured using projection Moiré on the mold surface and compared to finite element simulations. The elastic modulus and CTE of the substrate were measured using Stereo-DIC. However, the elastic modulus of the mold was measured using nano-indentation at different temperatures. A FE modelling and analytical equations based on the classical bending theory, along with the measured material properties, were employed to analyze the thermally induced deformations of the packages. A good agreement is achieved between FE and experiment. Furthermore, some fluctuations were observed in warpage measurements from package to package, which are consistent with the variability in the Young's modulus measurements of the MC material presented in the material's characterization section. The validated model can be used in the future to mitigate the warpage FCBGA packages by conducting a design of experiment.

As temperature changes homogeneously within the FCBGA package, different thermal strains are generated in the package resulting from the

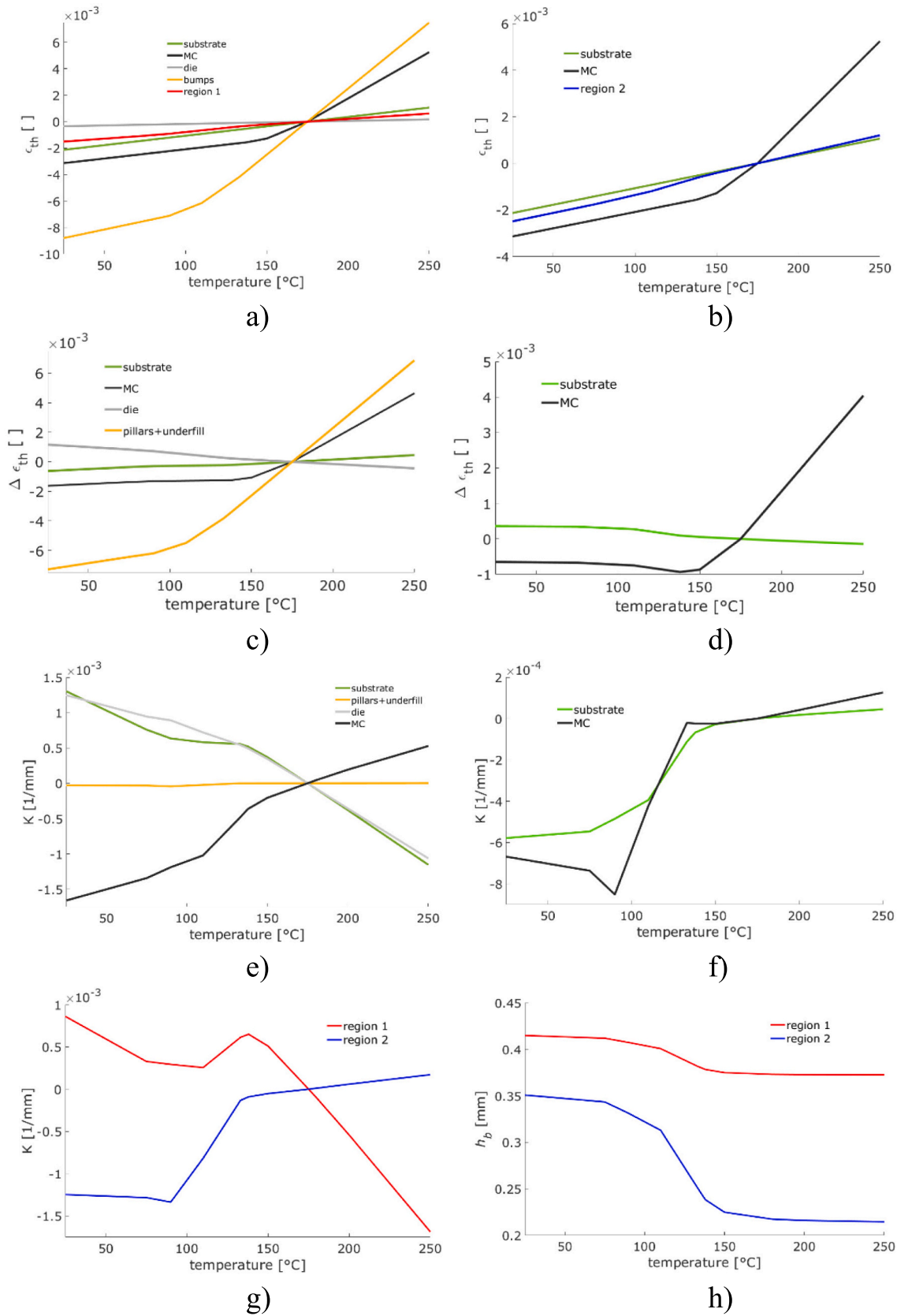


Fig. 10. Analytical calculation for the analysis of the FCBGA package warpage: **a-b** thermal strains in regions 1 and 2 respectively as well as their different constituent layers. **c-d** thermal strain mismatches between the different constituent layers of the stack and its average thermal strain for regions 1 and 2 respectively. **e-f** contribution of the different constituent layers to the overall curvature of the FCBGA package K in region 1 and 2 respectively. **g** overall curvature of region 1 and 2. **h** neutral bending plane height h_b .

CTE mismatch between the constituent material layers. Because of the difference in cross-sectional area within the package, it exhibits a different curvature in the die region at the center of the package compared to the region outside the die. At high temperatures, a concave or smiling warpage shape was shown in the die region while a convex or crying shape in the region outside the die. In contrast, at room temperature, the die area has a convex shape warpage, while outside this area presented a concave shape. According to the analytical equations, the pillars and underfill has less influence on the warpage compared to the die, the substrate and the mold material. However, the BEOL has no influence on the warpage because of its very low thickness which is around 3 μm .

The variation of elastic modulus and CTE of the MC material is the key parameter changing the warpage behavior with temperature variation, especially at the glass transition. In the die region, two bending moments counteracting each other, the bending moment created by the MC in one hand, and the one created by the die and substrate on the other hand. However, in the region outside the die, the two bending moments induced by the MC and substrate are in the same direction inducing a curvature in the opposite direction compared to the bending curvature in the die region.

CRedit authorship contribution statement

A. Salahouelhadj: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing, Visualization. **M. Gonzalez:** Conceptualization, Supervision, Writing – review & editing, Project administration. **K. Vanstreels:** Software, Validation, Investigation, Validation, Writing – review & editing. **G. Van der Plas:** Supervision, Writing – review & editing, Project administration. **G. Beyer:** Supervision, Project administration. **E. Beyne:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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